

Exercise 2

Assume that the signal $s(t)$ has a spectral density of energy

$$G(f) = 4b^2 \text{sinc}^2(fa) \cos^2(2\pi fa)$$

Calculate the autocorrelation of $s(t)$. 1

Furthermore, knowing that the phase of the transform of $s(t)$ is everywhere zero, exploit this result to derive the original expression of the signal $s(t)$. 2

a

$$\begin{aligned} G(f) &= 4b^2 \text{sinc}^2(fa) \cos^2(2\pi fa) \\ &= 4b^2 \text{sinc}^2(fa) \frac{1}{2} (1 + \cos(4\pi fa)) \\ &= 2b^2 \text{sinc}^2(fa) + 2b^2 \text{sinc}^2(fa) \cos(4\pi fa) \\ &= 2b^2 \text{sinc}^2(fa) + \cancel{2}b^2 \text{sinc}^2(fa) \frac{1}{\cancel{2}} (e^{j4\pi fa} + e^{-j4\pi fa}) \\ &= \underbrace{2b^2 \text{sinc}^2(fa)}_{\text{a}} + \underbrace{b^2 \text{sinc}^2(fa)}_{\text{b}} e^{j4\pi fa} + \underbrace{b^2 \text{sinc}^2(fa)}_{\text{c}} e^{-j4\pi fa} \end{aligned}$$

$\bullet \cos^2(x) = \frac{1}{2}(1 + \cos(2x))$

$\bullet \cos(x) = \frac{e^{jx} + e^{-jx}}{2}$

a

$$\begin{aligned} G_a(f) &= 2b^2 \text{sinc}^2(fa) \\ \omega &= \frac{\text{BASE}}{2} = \frac{\tau}{2} \rightarrow \tau = 2\omega \\ 2b^2 &= A \frac{\text{BASE}}{2} = A \frac{\tau}{2} \rightarrow 2b^2 = A \frac{2\omega}{2} \rightarrow A = \frac{2b^2}{\omega} \end{aligned}$$

$$\rightarrow F(G_a(f)) = \frac{2b^2}{\omega} \text{TRIANG}\left(\frac{t-0}{2\omega}\right)$$

NO DELAY

b

$$\begin{aligned} G_b(f) &= b^2 \text{sinc}^2(fa) e^{j4\pi fa} \\ \omega &= \frac{\text{BASE}}{2} = \frac{\tau}{2} \rightarrow \tau = 2\omega \\ b^2 &= A \frac{\text{BASE}}{2} = A \frac{\tau}{2} \rightarrow b^2 = A \frac{2\omega}{2} \rightarrow A = \frac{b^2}{\omega} \\ e^{j4\pi fa} &= e^{-j2\pi f \text{ PEAK-DELAY}} = e^{-j2\pi f t_1} \rightarrow \cancel{j4\pi fa} = -j2\pi f t_1 \rightarrow t_1 = -2\omega \end{aligned}$$

$$\rightarrow F(G_b(f)) = \frac{b^2}{\omega} \text{TRIANG}\left(\frac{t - (-2\omega)}{2\omega}\right)$$

c

$$\begin{aligned} G_c(f) &= b^2 \text{sinc}^2(fa) e^{-j4\pi fa} \\ \text{ALREADY DID FOR } \text{b} \end{aligned}$$

$$\rightarrow F(G_c(f)) = \frac{b^2}{\omega} \text{TRIANG}\left(\frac{t - 2\omega}{2\omega}\right)$$

PLOT

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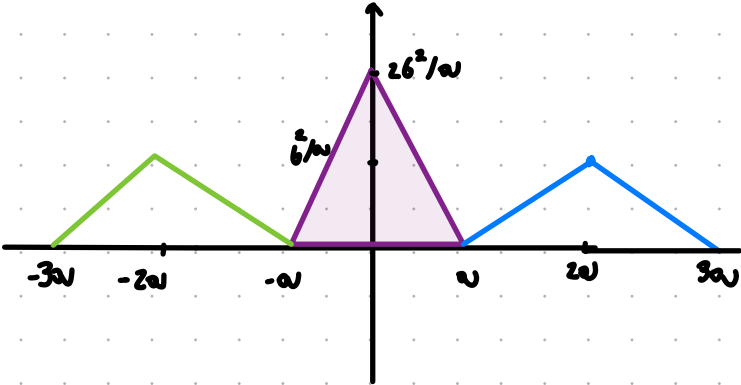
$$F(G_a(f)) = \frac{2b^2}{\omega} \text{TRIANG}\left(\frac{t-0}{2\omega}\right)$$

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$$F(G_b(f)) = \frac{b^2}{\omega} \text{TRIANG}\left(\frac{t - (-2\omega)}{2\omega}\right)$$

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$$F(G_c(f)) = \frac{b^2}{\omega} \text{TRIANG}\left(\frac{t - 2\omega}{2\omega}\right)$$



$R_s(\tau) = \text{a} + \text{b} + \text{c}$

2

$$\begin{aligned} G(f) &= |S(f)|^2 \xrightarrow{\text{if } \angle S(f) = 0} S(f) = \sqrt{G(f)} \\ S(f) &= \sqrt{G(f)} = 2b \text{sinc}(fa) \cos(2\pi fa) \\ &= 2b \text{sinc}(fa) \frac{1}{2} (e^{j2\pi fa} + e^{-j2\pi fa}) \\ &= \underbrace{b \text{sinc}(fa)}_{\text{a}} e^{j2\pi fa} + \underbrace{b \text{sinc}(fa)}_{\text{b}} e^{-j2\pi fa} \end{aligned}$$

$\bullet \cos(x) = \frac{e^{jx} + e^{-jx}}{2}$

a

$$\begin{aligned} \text{RECT: } \omega &= \text{BASE} = \tau \\ b &= A \text{BASE} = A\tau \rightarrow b = A\omega \rightarrow A = b/\omega \\ e^{j2\pi fa} &= e^{-j2\pi f \text{ PEAK-DELAY}} = e^{-j2\pi f t_1} \rightarrow \omega \dots t_1 \rightarrow t_1 = -\omega \end{aligned}$$

$$\rightarrow S_a(t) = \frac{b}{\omega} \text{RECT}\left(\frac{t - (-\omega)}{\omega}\right)$$

b

$$\begin{aligned} S_b(t) &= \frac{b}{\omega} \text{RECT}\left(\frac{t - \omega}{\omega}\right) \\ \text{SIMILAR TO } \text{a} \text{ BUT } t_1 = \omega \end{aligned}$$

$$S(t) = S_a(t) + S_b(t)$$